

Maximum Marks: 100

Total Time: 180 min

EE323-Analog and Digital Communication (EE-1, 2, 3)

FINAL EXAM (FALL 2015)

Dated: December 8, 2015

INSTRUCTIONS: Do not ask any questions from the invigilator; make your assumptions in case of confusion. Make neat and labeled figures/block diagrams where required. There are total six (6) questions and 4 pages. Solve all the parts of a question together otherwise they will not be marked.

Question #1 [7+8+5 = 20 marks]

- (a) Find the complex (exponential) Fourier series for the following signal $x(t)$ using Euler's formula. Then, find its Fourier transform and plot $X(\omega)$.

$$x(t) = \cos \omega_o t + \sin^2 \omega_o t$$

- (b) The spectrum of the message signal $m(t)$ is shown in Fig. 1 below. To ensure communication privacy, this signal is applied to a system (known as **scrambler**) shown in Fig 2 below. Analyze the system and sketch the spectrum at points A, B, C and the output $x(t)$. Properly label the axis information.

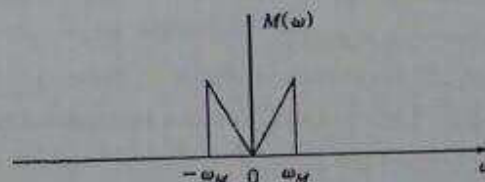


Figure 1 Spectrum of $m(t)$

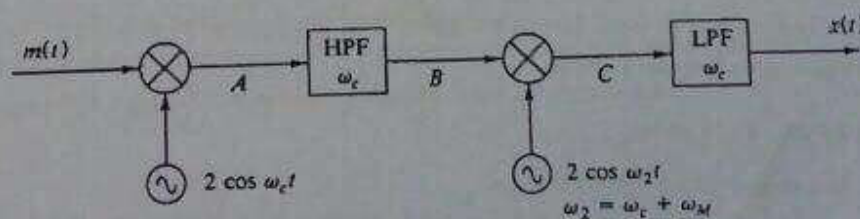


Figure 2 Scrambler

- (c) Derive the envelope detector condition for SSB signal with carrier (SSB+C). For derivation, use LSB.

Question # 2 [4+6+5 = 15 marks]

- (a) Draw the block diagram structures for the Transmitter and Receiver for VSB transmission.
- (b) The vestigial filter transfer function $H_i(\omega)$ of the transmitter module is shown below. The carrier frequency is $f_c = 10$ kHz and the baseband signal bandwidth is 4 kHz. Plot the corresponding transfer function of the equalizer filter $H_o(\omega)$ of the receiver module on the same scale as $H_i(\omega)$.

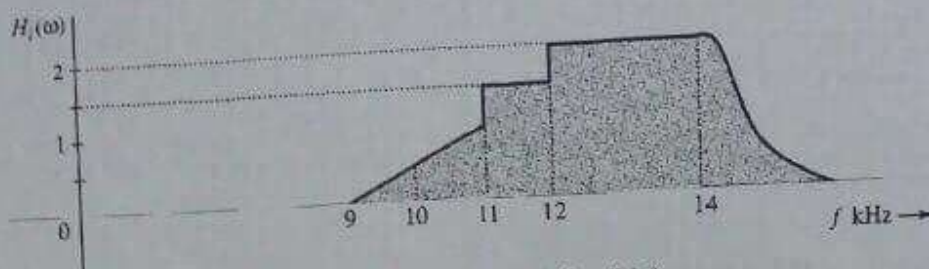


Figure 3 Spectrum of Vestigial filter $H_i(\omega)$

- (c) Mathematically prove that the Hilbert Transform is equivalent to a 90 degrees phase shifter.

Question # 3 [8+7 = 15 marks]

- (a) An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 10^6$ is described by

$$\varphi_{EM}(t) = 10 \cos(\omega_c t + 0.1 \sin(10^3) \pi t)$$

- Find the power of the modulated signal.
 - Find the frequency deviation Δf .
 - Find the phase deviation $\Delta\phi$.
 - Estimate the bandwidth of $\varphi_{EM}(t)$.
 - Considering $\varphi_{EM}(t)$ as a PM signal with $k_p = 10$, find $m(t)$.
 - Considering $\varphi_{EM}(t)$ as an FM signal with $k_f = 10\pi$, find $m(t)$.
- (b) Design an Armstrong indirect FM modulator to generate an FM carrier with a carrier frequency of 96 MHz and $\Delta f = 200$ kHz. A narrow-band FM generator with $f_c = 200$ kHz and adjustable Δf in the range of 9 to 10 Hz is available. The stock room also has an oscillator with adjustable frequency in the range of 9 to 10 MHz. There is a bandpass filter with any required center frequency, and only frequency doublers are available.

Question # 4 [6+9 = 15 marks]

- (a) State and prove the Sampling theorem.
- (b) A compact disc (CD) recording system samples each of two stereo signals with a 16-bit analog-to-digital converter (ADC) at 44.1 kbps.
- Determine the output signal-to-quantizing-noise ratio for a full-scale sinusoid.

- (ii) The bit stream of digitized data is augmented by the addition of error-correcting bits, clock extraction bits, and display and control bit fields. These additional bits represent 100 percent overhead. Determine the output bit rate of the CD recording system.
- (iii) The CD can record an hour's worth of audio file. Determine the number of bits recorded on a CD.

Question # 5 [10+5 = 15 marks]

- (a) A message signal $m(t)$ is transmitted by binary PCM without compression. If the SNR (signal-to-quantization-noise ratio) is required to be at least $\left(\frac{s_o}{N_o}\right)_{dB} = 50$ dB, determine the minimum value of L required for the following message signals. Also determine the SNR obtained with this minimum L for these message signals.

(i) $m(t)$ is sinusoidal with peak amplitude m_p .

(ii) $m(t)$ is triangular with peak amplitude m_p .

- (b) For a binary PCM signal, determine L if the compression parameter $\mu = 100$ and the minimum SNR required is 50 dB. Determine the output SNR with this value of L .

Hint: $\frac{s_o}{N_o} = 3L^2 \left(\frac{m^2(t)}{m_p^2} \right)$ changes to $\frac{s_o}{N_o} = \frac{3L^2}{(\ln(1+\mu))^2}$ for μ -law compander.

Question # 6 [5+7+8 = 20 marks]

- (a) Draw the **essential** block diagram of a typical digital communication system (DCS) starting from information source upto the sink with proper labeling.

- (b) Explain the basis for deriving the likelihood ratio test

$$z(T) \underset{H_2}{\overset{H_1}{\gtrless}} \gamma \quad \frac{q_1 + q_2}{2}$$

and under what conditions the above test would be the optimum decision making criteria.

- (c) A matched filter is a linear filter designed to provide the maximum SNR at its output for a given transmitted symbol waveform. SNR at sampling instant $t = T$, is

$$\left(\frac{S}{N} \right)_T = \frac{a_1^2}{\sigma_0^2} \left\{ \frac{E_b}{N_0} \right\}$$

Find the filter transfer function $H(f)$ and impulse response $h(t)$ of the filter that maximizes the above SNR equation. Also comment on the dependency of maximum output SNR.

PSD, not

PL 22
R 22
50
11
5